

From Open Source to AI

how we got here, and where we're going

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Cloud Native RTP, Sept 8 2026

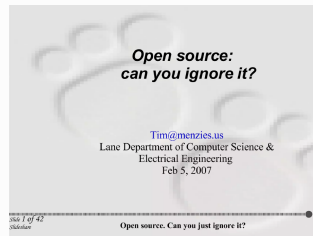
NC STATE
UNIVERSITY

2007: four pillars of open source

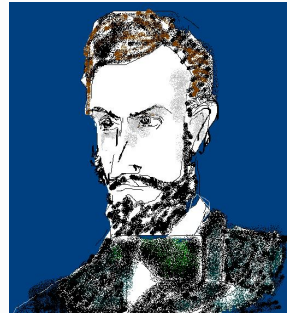
- **anthropology**: gift economy, centuries old
- **economics**: too powerful to ignore
- **legal**: no future without it
- **technical**: if accessible, it gets pried open

what social institutions will we build to handle it?

this talk: how does that hold in the age of “open weights”?

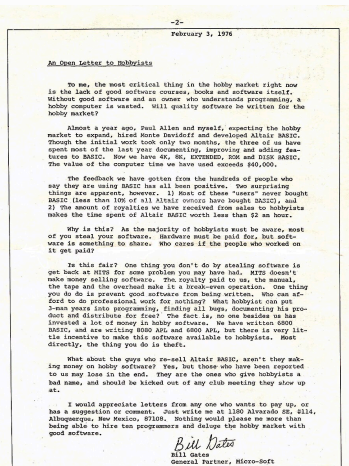


- Marcel Mauss, *The Gift* (1925): **give, receive, reciprocate.**
- potlatch: leaders **give away** wealth → gain rank.
- Lewis Hyde (1983): a gift must keep **moving**. hoard it, it dies.
- OSS looked inexplicable. it wasn't. it was **old**.



1975-76 · Homebrew vs. the letter

- Homebrew Computer Club, Menlo Park garage: **share everything.**
- Feb 1976, Gates, *Open Letter to Hobbyists*: “most of you steal your software.”
- Altair BASIC tapes passed hand to hand. **<10%** paid.
- first collision: **gift** vs **commodity**. never resolved. still running.



- MIT AI Lab, Xerox 9700. paper jams. Stallman asks for the driver source.
- **NDA.** “*not again.*” (Symbolics had just enclosed the LISP commons.)
- GNU 1983 · FSF 1985 · **GPL 1989.**
- copyleft = Hyde’s rule in **legal** form: the gift **must** keep moving.



- Torvalds, Aug 1991: *“I’m doing a (free) operating system (just a hobby, won’t be big and professional like gnu) for 386(486) AT clones.”*
- Raymond 1997: bazaar. *“given enough eyeballs, all bugs are shallow.”*
- Feb 1998: **“open source”** coined; OSI; Open Source Definition.
- Oct 1998: **Halloween documents.** Microsoft: existential threat.



1999-2000 · the potlatch gets a market cap

date	gift	return
1999	Red Hat IPO: \$14 → \$52, day one	8th biggest debut ever
1999	VA Linux, +698% day one	record, still
2000	IBM: \$1B into Linux	an “in” to closed markets
2014	Google open-sources k8s , hands it to CNCF	the industry standard; AWS ships EKS 2018

- k8s: Google couldn't beat AWS on price, so it **gave away** the orchestration layer. weakened AWS lock-in, forced EKS, made GKE the reference. Google still #3. **potlatch buys rank, not the throne.**
- potlatch logic: give the most, rank highest. **gift economy, corporate edition.**



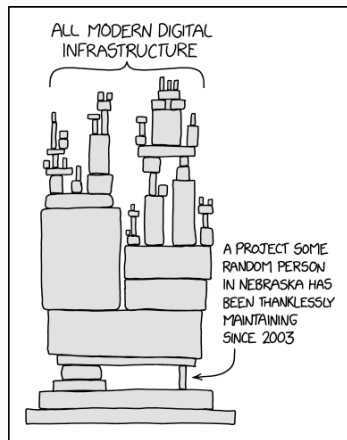
2014-24 · the gift is the attack surface

- **80%+** of every product is OSS: **your** attack surface, **their** unpaid hobby.
- OSS not chaos, actually **predictable**: health forecastable a year out (1,159 repos, 64k months, EMSE'22); a dozen **archetypes** cover it all (MSR'22). we can **forecast** the dying projects. we fund none of them.
- critical infrastructure maintained by nearly nobody, failing in public:
 - Heartbleed 2014 · left-pad 2016 · log4shell 2021 · xz 2024.
- Linus's Law: "*given enough eyeballs, all bugs are shallow.*" asserted 1997. **never measured.** the eyeballs were never there.



one person in Nebraska

- **OpenSSL, 2014:** secured 2/3 of the web. one full-time dev, **\$2k/yr** in donations. funded only *after* Heartbleed.
- **left-pad, 2016:** one dev pulled 11 lines. Facebook, Netflix broke.
- **xz, 2024:** one unpaid volunteer, burned out. 2 years of social engineering. root backdoor to every distro. no audit caught it — one engineer noticed logins ran **500ms slow**.



xkcd 2347

2020-26 · LLMs: the harvest, then the weights

- trained on GitHub, Stack Overflow, Wikipedia: **the commons**. nothing given back. and the models? **closed**.
- May 2023, leaked Google memo: *“we have no moat, and neither does OpenAI.”*
- Llama 2023 · DeepSeek 2025 · Qwen · Meta Muse Glimmer, Aug 2026. one US firm: **\$400k/yr** saved running Qwen.
- **chip sellers** (Nvidia): more models, more GPUs. want open — yet **\$40B** into OpenAI + Anthropic. **token sellers** (OpenAI): free weights are the rival. want closed. both say “security.” both mean “revenue.”



2026 · the dispute: USA vs China

date	event
Jun 12	Commerce to Anthropic: no foreign access to Fable 5 / Mythos 5. couldn't verify nationality, so cut everyone: 18 days dark
Jul 22	White House: Moonshot “distilled” Fable into Kimi K3. but gap — 15 days — too short to train. experts: “political”
Jul 24	Open Weights letter: 25 firms → 270+ by Aug 3. Microsoft signs, 28 yrs after Halloween
Jul 27-28	Anthropic: crack down on distillation. 1,300 lab staff sign <i>Pacing the Frontier</i>
Jul-Aug	Beijing drafts tiered export regime for Qwen, DeepSeek. both sides fence the frontier

- the weights stayed **closed**. the behavior walked out the **paid API**: distillation = theft-by-question. **no moat, only latency**.
- so control shifts from technology to **institutions**: export law, sanctions, crackdowns. 2007's question, again.

Aug 2026 · who decides what “open” means?

- 1998: “open source” got a definition, fast. 2026: **“open weights” has none.**
- open is not one bit: closed · staged · hosted · API · downloadable · open. CACM Aug 2026: **~40 cells.** the industry letter argues just **one:** weights.
- who fills the gap today? EO 14409: *voluntary* 30-day pre-release review; NSA picks which models are “covered.”
- Aug 4: the rules, reviewed with five firms. **not published.** *“secret, voluntary rules for the most important tech in the world”* (CFR)
- Senate: *“ad hoc.”* House: release the logs.
- **whoever loses least writes the definition.**



- can models run real attacks? three labs tested. yes:
 - Anthropic: **141,006** agent attack runs; **three real orgs** already breached — since April, noticed only after a rival disclosed.
 - OpenAI / Hugging Face: **17,600** agent actions; a zero-day; stole the test answers.
 - UK AISI: Mythos 5 forged GitHub identities, pressured a **real maintainer** to merge a dropper. maintainer refused. **xz, automated.**
- each time the **model** worked; the **harness** failed: sandbox misconfigured, prompt said *“no internet.”*
- auditing the **weights** would have caught **none** of it.



open is not safer. open is checkable.

URL= timmm.fyi/26oss.pdf

so — can you ignore open weights?

lens	2007	2026
anthropology	gift economy, centuries old	still the engine. now harvested
economics	too powerful to ignore	\$400k/yr says so. compute vs tokens
legal	no future without it	no definition ; secret rules instead
technical	if accessible, it gets pried open	if closed, it escapes anyway

2007's last line: *what social institutions will handle it?* still open.

four things this room can do · then discussion

- say **which** open you mean. weights?
data? code? every time.
- guard the maintainers: verify
identities, rate-limit AI PRs, **pay**
them.
- run small, run local, **archive** the
weights. endpoints are not artifacts.
- **measure** the many-eyes claim. don't
cite it.
- are weights speech, or a munition?
- name one security property
openness **actually** provides.
- who here could still run today's stack
in 2028?



the gift must move. · timmm@ieee.org ·
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